PRODUCT DESIGN & DEVELOPMENT ENGINEER

Julian **SCHAPINK**

As an INSA Lyon engineer, I led three end-to-end R&D projects at NTN-SNR Roulements: full mechatronic design, multiphysics FE simulation, and additive manufacturing. A deliberate detour through École 42 and co-founding a startup taught me to think product as much as part. I am looking for a product design engineering role within a company building ambitious products.

(+33)6 20 09 11 06 | Lyon (FR) | Licence | julian.schapink@gmail.com | jschapink.github.io

TRAININGS

Master's in Mechanical Engineering
INSA Lyon
SEPT. 2017 - SEPT. 2021
Product Design and Innovation
(work-study programme)

Mechanical and Production DUT
IUT Lyon 1
SEPT. 2015 - SEPT. 2017
Design, Industrialisation and
Industrial Organisation
(semester and internship in Quebec)

Digital Architect
École 42 Paris
AUG. 2022 - DEC. 2023
Software architecture
C developpement, bash, python

CPGE Martinière Monplaisir
PTSI Monplaisir Lyon
SEPT. 2014 - SEPT. 2015
Physics, Technology and Engineering Sciences

SKILLS

Design
CAD SolidWorks, GD&T,
DFMA, Additive manufacturing

Simulation
Multiphysics FEA, Forge NXT,
Topology optimisation

Methods
Taguchi design of experiments,
Matlab, ISO standard Excel VBA

Systems
Mechatronics, Instrumentation,
Electro-pneumatics, Arduino/C

Languages
Bilingual French-English (TOEIC 980/990,
TOEFL 105/120), Spanish B2

EXPERIENCES

CINEFLOWS — JAN. 2024 - SEPT. 2024, PARIS

Co-founder, Product Manager

- Development of a B2B SaaS solution for film distribution:

  → team coordination, needs analysis, definition of functional specifications, value proposition assessment, UX Design training, web-app prototyping.

NTN-SNR ROULEMENTS — SEPT. 2017 - SEPT. 2020, ANNECY

New Process Development Engineer

- Developed a high-throughput autonomous sorting machine for tapered rollers (350 pcs/min, $\Delta\varnothing < 3$ mm):

  → led end to end: analysis, technology selection, mechanical design and electro-pneumatic circuits, dedicated kinematics, embedded Arduino/C control with Keyence laser sensors, commissioning, on-site testing, process validation and handover to production engineering.

- Redesigned an aeronautical engine bearing cage using metal additive manufacturing (MAM):

  → topology optimisation (SolidWorks FE), mass reduction, integration of emergency lubrication channels impossible to machine conventionally, prototyping. MAM exploration to motivate R&D investment in the new process.

- Optimised the localised induction heat-treatment (LHT) process:

  → Taguchi design of experiments (mechanics, thermal physics, electromagnetism, materials science), sensitivity analysis of FE simulation parameters, test inductors made by copper AM, test campaigns with Forge NXT editor for solver calibration. Simulation management and data setup, VBA parameterisation tools delivered and in use by the LHT department; development cycle made more efficient.

INTERESTS

Framebuilding | Prototyping, DIY and Maker culture | IT and electronics | Car mechanics
Interior architecture and renovation | Photography | Industrial design | Cycling, alpine skiing and hiking